A Symptom-Based Multi-Disease Prediction System Using Machine Learning

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Abstract: This project develops a multi-disease prediction system using machine learning for early disease identification based on patient symptoms. Unlike traditional models that detect only one disease, this system predicts multiple conditions such as Diabetes, Asthma, Jaundice, Tuberculosis, Pneumonia, and Heart-related diseases through a single platform. The dataset is pre-processed and trained using algorithms like Logistic Regression, SVM, Decision Tree, Random Forest, XGBoost, and ANN to find the most effective model. Among these, ensemble models like Random Forest, XGBoost and AdaBoost provide better performance and reliable predictions. The system is developed using Python for implementation and trained on a structured symptom-based dataset for multi-class disease classification. The system enables quick and accurate preliminary diagnosis and demonstrates how machine learning can support medical professionals in better decision-making and early care. The system also evaluates model performance using accuracy, precision, recall, and F1-score selecting the most efficient prediction system.

I INTRODUCTION

Healthcare systems today are increasingly adopting machine learning technologies to improve early disease identification and medical diagnosis. Many life-threatening diseases such as diabetes, heart disease, tuberculosis, pneumonia, and jaundice continue to affect a large population worldwide, especially when they are not identified at an early stage. Conventional diagnostic procedures often depend on manual evaluation and clinical experience, which requires significant time and may lead to delays in treatment. With the growth of digital healthcare data and intelligent computing, machine learning techniques have created new opportunities to analyze symptoms and predict diseases more accurately and efficiently.

Recent developments in artificial intelligence and data analysis have enabled the use of classification algorithms to support medical decision-making. Machine learning models can process large symptom-based datasets and determine hidden patterns that might not be easily observed by humans. Different computational models are widely utilized to analyze symptom data and identify possible diseases. Approaches including Logistic Regression, Decision Tree, Support Vector Machine, Random Forest, and boosting algorithms have shown promising results in predicting diseases based on symptoms and patient data. These techniques enhance accuracy of diagnosis and lower human errors while supporting medical professionals in making faster decisions.

However, many existing systems are designed to predict only a single disease, limiting their usefulness in real-world healthcare applications. Patients often show symptoms that may be related to multiple conditions, making it necessary to develop systems capable of predicting more than

one disease at a time. Nevertheless, limited research focuses on integrated multi-disease prediction using ensemble learning techniques within a unified framework. This highlights the need for an integrated and intelligent multi-disease prediction system that can analyze symptoms collectively and provide accurate predictions through a single platform.

Based on these advancements, this project proposes a multi-disease prediction platform capable of identifying several diseases using a symptom-based dataset. The system integrates multiple predictive algorithms within a unified framework to improve performance and efficiency. By implementing and comparing different computational models, the system aims to provide reliable predictions and assist in early disease detection. The proposed approach demonstrates how data-driven techniques can be utilized to develop effective healthcare support systems and enhance the overall efficiency of medical diagnosis.

II LITERATURE SURVEY

Kumar and Khan (2025) proposed a machine learning-based approach that integrates various health conditions through a single framework and presented classification results using publicly available datasets through a user-friendly interface. Their study demonstrated reduced manual effort and improved predictive capability; however, it primarily relied on conventional machine learning algorithms and did not explore advanced ensemble techniques or multi-class integrated learning approaches.

Ramesh et al. (2023) developed a feasible multi-disease prediction model using symptoms and clinical features. Their work highlighted the importance of predictive modelling for early diagnosis and accessibility. Nevertheless, the study mainly focused on traditional machine learning techniques and did not investigate advanced methods such as model fusion or interpretability techniques.

Din et al. (2024) introduced a multi-disease prediction framework emphasizing the role of preprocessing and comparative evaluation of classifiers. Although the results were promising, the research concentrated mainly on algorithm comparison and did not implement a unified real-time system capable of consolidating multiple diseases under a single structured framework.

Shaikh et al. (2021) presented a web-based multi-disease prediction application that leveraged computational models to estimate disease risk. While the study demonstrated the practical usability of an application-based system, it did not extensively examine advanced boosting or scalable ensemble methods.

Mathew et al. (2024) presented a multi-disease prediction model trained on publicly available datasets to diagnose various health conditions. Although the platform provided a cohesive interface for multiple disease detection, it relied largely on standard supervised learning techniques and did not fully address complex multi-class symptom relationships.

Sharma et al. (2024) developed a multi-disease detection system using a structured symptom dataset. Their work demonstrated the effectiveness of integrating predictive models with user-friendly interfaces to assist in early diagnosis. Despite these contributions, the framework mainly relied on traditional classification approaches and did not investigate advanced model combination methods or optimization strategies to improve reliability and performance.

III PROPOSED METHODOLOGY

A Data Collection

The model initially uses the collection of a structured symptom-based medical dataset stored in CSV format. The dataset contains patient health records with multiple symptom features and corresponding disease labels. Both training and testing datasets are utilized to assist model

development and performance assessment. This structured dataset serves as the basis for developing a reliable multi-disease prediction system by facilitating the analysis of symptom-related data.

B Data Preprocessing

Data preprocessing focuses on cleaning and preparing the dataset to ensure accuracy and consistency. Relevant diseases are selected, unnecessary columns are removed, and missing values are handled. Symptom data is converted into binary format to represent presence or absence, ensuring reliable input for model training.

The overall proposed system architecture is depicted in Figure 1.

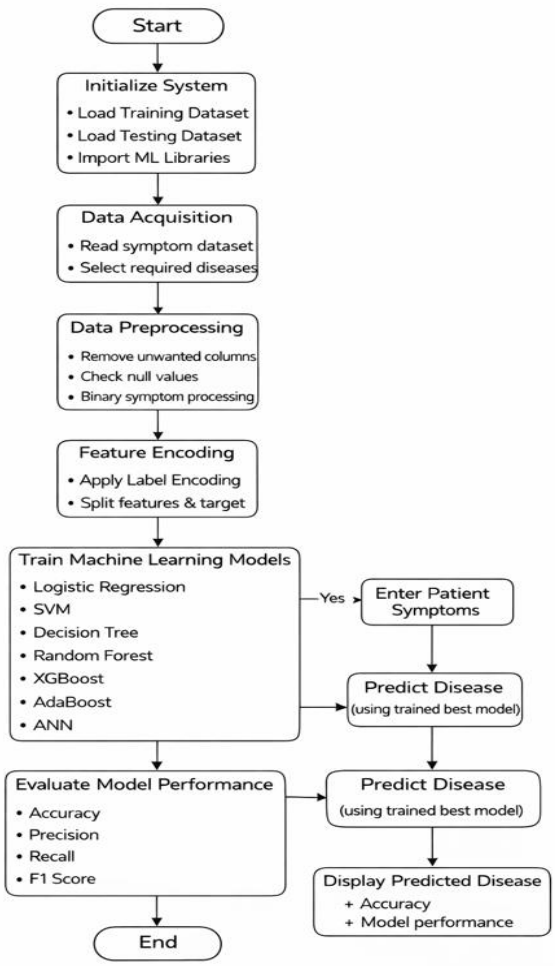

Figure 1: Proposed System Architecture.

C Feature Encoding

At this step, disease categories that are represented in text form are converted into numerical values through a label encoding approach. After this conversion, the dataset is arranged into input features and their respective target outputs. The data is then partitioned into training and evaluation sets to assess the effectiveness of the prediction models and control overfitting. This preparation ensures that the dataset is organized and ready for further analysis.

D Model Training

The prepared dataset is used to train multiple classification models that analyze patterns between symptoms and disease conditions. The implemented algorithms include Logistic Regression, Support Vector Machine, Decision Tree, Random Forest, XGBoost, AdaBoost, and Artificial Neural Networks. During training, these models study the relationships between symptom inputs and the related disease labels to improve prediction capability.

E Model Performance Evaluation

After completing model training, their predictive capability is assessed using evaluation metrics such as accuracy, precision, recall, and the F1-score. These measures help analyze how accurately diseases can be identified from the given symptom data. A comparison of the results obtained from each model is then performed to identify the approach that delivers the most consistent and reliable predictions.

F Best Model Selection

According to the performance results, the best optimal model is chosen for implementation. The selection relies on overall performance across multiple evaluation metrics. This chosen model becomes the final prediction model used by the system for disease classification and real-time symptom-driven diagnosis in practical healthcare scenarios.

G Disease Prediction

In the prediction stage, the user provides symptom inputs to the system. These inputs are processed and passed to the trained best-performing model, which analyzes the symptom patterns and predicts the most likely disease. The prediction is generated quickly, enabling efficient and automated disease identification.

IV MODELS

A Logistic Regression

To identify possible diseases from symptom inputs, a statistical prediction method is applied that calculates the probability of each disease category based on the available patient information. The method studies how symptom features are associated with the corresponding disease category in the dataset. By analyzing these relationships, the model helps identify possible health conditions based on the symptom patterns observed in the data.

B Support Vector Machine (SVM)

Support Vector Machine is employed to categorize diseases by constructing a boundary that separates different classes within the dataset. This approach works well when dealing with datasets that contain many symptom features. In this study, the model contributes to distinguishing diseases that share similar symptoms and supports more accurate classification of patient conditions.

C Decision Tree

In this study, a tree-based classification technique is used to model decision rules through a hierarchical structure for effective disease prediction. The dataset is gradually partitioned into smaller segments based on important symptom attributes at each level of the tree structure. Such an organized structure makes the results easier to interpret and helps clearly explain how different symptom patterns relate to disease predictions.

D Random Forest

For disease classification, a tree-based ensemble model is employed to strengthen prediction accuracy. Several decision tree structures are trained using different samples of the dataset to capture diverse symptom patterns. The outcome is obtained by combining the predictions produced by these trees. This process improves the stability of predictions and reduces the chances of overfitting.

E AdaBoost

AdaBoost, or Adaptive Boosting, is a boosting approach that combines various simple models to create a stronger prediction system. During training, the method gives greater weight to samples that are misclassified so that the model can learn from its previous errors. Through repeated iterations, the algorithm gradually improves its ability to handle difficult cases and enhances overall prediction accuracy.

F XGBoost

XGBoost is an advanced gradient boosting algorithm known for its fast and high accurate results in classification tasks. Models are created sequentially, where each new model corrects earlier errors. In this system, XGBoost is used to improve prediction accuracy and handle complex symptom patterns efficiently.

G Artificial Neural Networks (ANN)

Neural network-based models are brain inspired models with interconnected neuron layers that learn complex data patterns. They capture intricate relationships in data. In this project, an ANN model is implemented to analyze symptom data and classify diseases by learning from multiple hidden layers.

V RESULTS AND EVALUATION

The developed multi-disease prediction system was implemented using several data-driven classification approaches to examine their

capability in identifying diseases based on symptoms. The dataset was first preprocessed and filtered to include major diseases such as Diabetes, Bronchial Asthma, Jaundice, Tuberculosis, Pneumonia, and heart-related conditions. Multiple predictive models were trained and evaluated using the prepared symptom dataset to analyze how effectively diseases could be identified from the given inputs.

The evaluation shows that most of the implemented predictive models performed efficiently due to the structured and symptom-oriented nature of the dataset. Some classification approaches demonstrated strong capability in distinguishing diseases from symptom patterns. Other models also showed competitive performance, while several tree-based approaches produced stable and consistent prediction outcomes. The neural-based model also demonstrated its capability to learn complex symptom relationships and generate reliable disease predictions.

Among all the evaluated techniques, model-combination approaches provided more consistent and dependable results because they can effectively manage multiple features and reduce the risk of overfitting. The system was also tested with custom symptom inputs to examine its practical usability, and it successfully predicted possible diseases based on the symptoms provided.

The performance comparison of all models is presented in Table 1.

Table 1: Overall Performance Metrics of the developed framework.

S. No.	Algorithm	Accuracy (%)
1	Logistic Regression	97.7%
2	SVM	97.7%
3	Decision Tree	93.8%
4	Random Forest	93.3%
5	ANN	95.0%
6	XGBoost	97.2%
7	AdaBoost	96.0%

Overall, the outcome demonstrates that machine learning models can be effectively utilized for early disease prediction and medical decision support. The developed system offers a simple, efficient, and reliable approach for multi-disease prediction, assisting healthcare professionals in preliminary diagnosis and improving the speed and quality of medical decision-making. Furthermore, integrating multiple classification algorithms within a unified framework enhances the robustness and adaptability of the system. Comparative analysis indicates that ensemble and boosting-based models deliver more consistent and stable performance than individual classifiers, primarily due to their ability to manage feature complexity and minimize overfitting.

VI CONCLUSION

This study introduces a symptom-oriented multi-disease prediction framework developed to recognize multiple health conditions from patient symptom information. The system focuses on supporting the timely identification of illnesses including Diabetes, Bronchial Asthma, Jaundice, Tuberculosis, Pneumonia, and heart-related problems, which may become serious if diagnosis is delayed. Different predictive modelling techniques were applied and analyzed to determine the most suitable approach for identifying diseases based on symptom patterns.

The experimental results indicate that model-combination techniques produced higher accuracy and more stable performance compared with individual models. The developed system can predict diseases based on user-provided symptoms and emphasizes the significance of data-driven technologies in supporting healthcare applications. Overall, the proposed framework provides a fast and reliable approach for preliminary disease assessment and can help healthcare providers make better treatment decisions. The study also demonstrates how intelligent computational systems can contribute to future healthcare environments by enabling faster and more dependable disease prediction.

VII FUTURE WORK

The proposed multi-disease prediction system can be strengthened with a larger and more diverse dataset that includes additional diseases and real-time patient data to achieve more prediction accuracy and consistency. Future work can focus on integrating the system with a web or mobile applications to improve accessibility and make it more user-friendly for patients and healthcare professionals. Advanced deep learning techniques and real-time data processing can also improve model performance and enable continuous monitoring of patient health conditions. In addition, the system can be integrated with Internet of Things (IoT)-based wearable devices to collect live health parameters for accurate and timely predictions. Further improvements may include deploying the model on cloud platforms to enable scalable and secure access for healthcare institutions and support large-scale medical data analysis.

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